#13 二重积分换序与换系

类型一 直角坐标换序

【例 1】积分 $I=\int_0^{\ln 2}\mathrm{d}x\int_{\mathrm{e}^x}^2\frac{\mathrm{e}^{xy}}{y^y-1}\mathrm{d}y=$_____.

【例 2】积分 $\int_0^1\mathrm{d}y\int_0^{1-\sqrt{2y-y^2}}\frac{\mathrm{d}x}{\sqrt{2-x}}=$_____.

类型二 直角坐标换极坐标

【例 3】设 D：$2x\leqslant x^2+y^2$，$0\leqslant y\leqslant x\leqslant 2$，则 $I=\iint\limits_D\frac{\mathrm{d}x\mathrm{d}y}{\sqrt{x^2+y^2}}=$_____.

类型三 极坐标换序、换系原理及易错点

【例 4】$I=\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{\frac{1}{\sin\theta}}f(r)r\mathrm{d}r=$ _____.

A. $\int_0^1\mathrm{d}x\int_0^1 f(\sqrt{x^2+y^2})\,\mathrm{d}y$.

B. $\int_0^1\mathrm{d}x\int_1^x f(\sqrt{x^2+y^2})\,\mathrm{d}y$.

C. $\int_0^1 \mathrm{d}r \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} f(r) r \,\mathrm{d}\theta + \int_1^{\sqrt{2}} \mathrm{d}r \int_{\frac{\pi}{4}}^{\arcsin\frac{1}{r}} f(r) r \,\mathrm{d}\theta$.　　D. $\int_0^{\sqrt{2}} \mathrm{d}r \int_{\arcsin\frac{1}{r}}^{\frac{\pi}{4}} f(r) r \,\mathrm{d}\theta$.

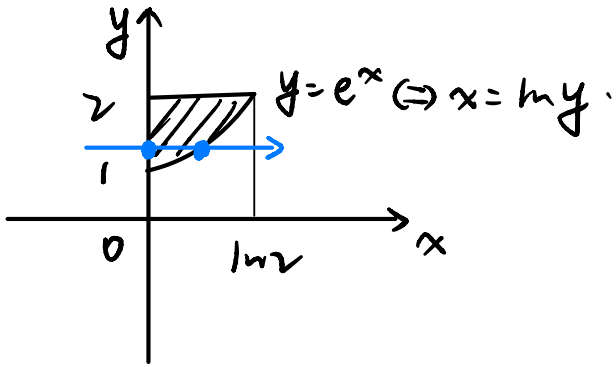
类型四 极坐标换序

【例 5】设 $D=\left\{(x,y)\middle| x^2+y^2\le 1, 0\le y\le x\right\}$，计算 $I=\iint\limits_D \frac{xy}{1+x^2-y^2}\mathrm{d}x\mathrm{d}y$.

类型五 极坐标换系

【例 6】$I=\int_{\arctan\frac{1}{4}}^{\frac{\pi}{4}} \mathrm{d}\theta \int_{\frac{1}{\sqrt{\sin\theta\cos\theta}}}^{\frac{2}{\cos\theta}} r^2\cos\theta \mathrm{d}r =$ _____.

【例1】积分 $I=\int_0^{\ln 2}\mathrm{d}x\int_{\mathrm{e}^x}^{2}\frac{\mathrm{e}^{xy}}{y^y-1}\mathrm{d}y=$_____.



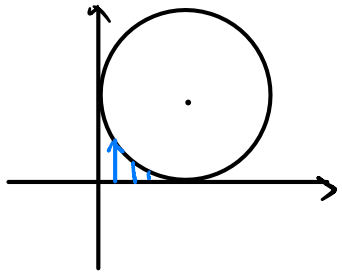
$$I=\int_1^2 dy\int_0^{\ln y}\frac{e^{xy}}{y^y-1}dx$$

$$=\int_1^2\left[\frac{1}{y}\frac{e^{xy}}{y^y-1}\right]_0^{\ln y}dy$$

$$=\int_1^2\frac{1}{y}\cdot\frac{e^{y\ln y}-1}{y^y-1}dy$$

$$=\ln y\Big|_1^2=\ln 2.$$

【例2】积分 $\int_0^1 dy\int_0^{1-\sqrt{2y-y^2}} \frac{dx}{\sqrt{2-x}} =$ _____.



$x = 1-\sqrt{2y-y^2}$，$(x-1)^2+(y-1)^2=1$

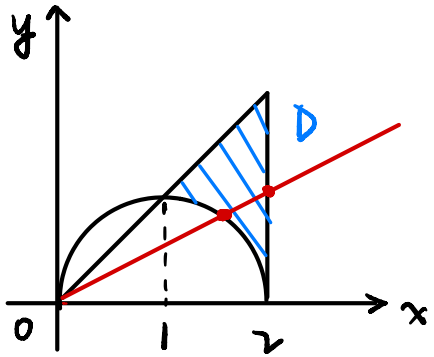
$I = \int_0^1 dx \int_0^{1-\sqrt{2x-x^2}} \frac{1}{\sqrt{2-x}} dy$

$= \int_0^1 \frac{1}{\sqrt{2-x}}(1-\sqrt{2x-x^2})dx$

$= \int_0^1 (\frac{1}{\sqrt{2-x}} - \sqrt{x})dx = (-2\sqrt{2-x} - \frac{2}{3}x^{\frac{3}{2}})\Big|_0^1$

$= -2-\frac{2}{3}-(-2\sqrt{2}) = 2\sqrt{2}-\frac{8}{3}$

【例3】设 D：$2x \leqslant x^2+y^2$，$0 \leqslant y \leqslant x \leqslant 2$，则 $I=\iint\limits_D \dfrac{\mathrm{d}x\mathrm{d}y}{\sqrt{x^2+y^2}}=$_____.



显然，区域D适合采用极坐标计算.

$$I=\iint_D \frac{dxdy}{\sqrt{x^2+y^2}}=\int_0^{\frac{\pi}{4}}d\theta\int_{2\cos\theta}^{2\sec\theta}\frac{1}{r}\cdot r\,dr=2\int_0^{\frac{\pi}{4}}(\sec\theta-\cos\theta)d\theta$$

$$=2(\ln|\sec\theta+\tan\theta|-\sin\theta)\Big|_0^{\frac{\pi}{4}}=2\ln(1+\sqrt{2})-\sqrt{2}$$

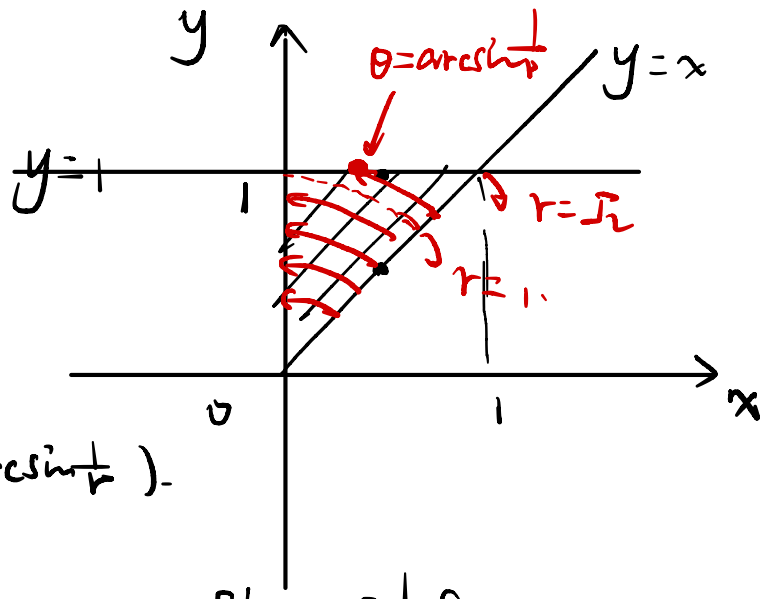
【例4】 $I=\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}d\theta\int_{0}^{\frac{1}{\sin\theta}}f(r)\,dr=$

A. $\int_0^1 dx\int_0^1 f(\sqrt{x^2+y^2})\,dy$.

B. $\int_0^1 dx\int_1^x f(\sqrt{x^2+y^2})\,dy$.

C. $\int_0^1 dr\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} f(r)\,r\,d\theta+\int_1^{\sqrt{2}} dr\int_{\frac{\pi}{4}}^{\arcsin\frac{1}{r}} f(r)\,r\,d\theta$.

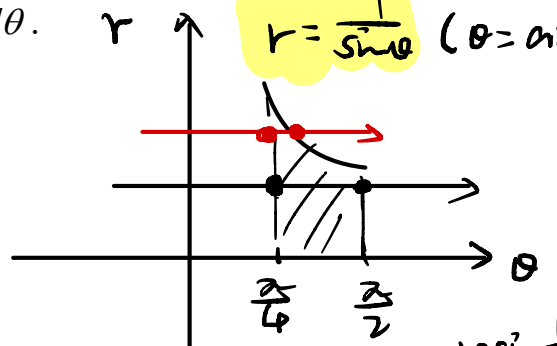
D. $\int_0^{\sqrt{2}} dr\int_{\arcsin\frac{1}{r}}^{\frac{\pi}{4}} f(r)\,r\,d\theta$.



$\int_0^1 dx\int_x^1 f(\sqrt{x^2+y^2})\,dy$

$y=r\sin\theta=1$

$r=\frac{1}{\sin\theta}$ $(\theta=\arcsin\frac{1}{r})$.

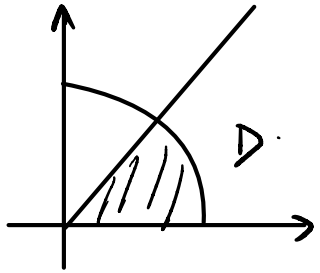


$\int_0^1 dr\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} f(r)\,r\,d\theta+\int_1^{\sqrt{2}} dr\int_{\frac{\pi}{4}}^{\arcsin\frac{1}{r}} f(r)\,r\,d\theta$

先 r 后 θ. 看 θ （画图解不等 r）.

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【例5】设 $D=\{(x,y)\mid x^2+y^2\le 1,0\le y\le x\}$，计算 $I=\iint\limits_D \frac{xy}{1+x^2-y^2}\mathrm{d}x\mathrm{d}y$.



$$I=\int_0^{\frac{\pi}{4}}d\theta\int_0^1\frac{r^2\cos\theta\sin\theta}{1+r^2(\cos^2\theta-\sin^2\theta)}r\,dr$$

$$I=\int_0^1 dr\int_0^{\frac{\pi}{4}}\frac{1}{2}\cdot\frac{r^3\sin 2\theta}{1+r^2\cos 2\theta}d\theta$$

$$=\int_0^1-\frac{1}{4}r\ln(1+r^2\cos 2\theta)\Big|_0^{\frac{\pi}{4}}dr$$

$$=\int_0^1\frac{1}{4}r\ln(1+r^2)dr$$

$$=\frac{1}{8}\int_0^1\ln(1+r^2)d(r^2+1)$$

$$=\frac{1}{8}(1+r^2)\ln(1+r^2)\Big|_0^1-\frac{1}{8}\int_0^1(r^2+1)\cdot\frac{2r}{1+r^2}dr$$

$$=\frac{1}{8}(2\ln 2-1).$$

$$[\ln(1+r^2\cos 2\theta)]'=\frac{-2r^2\sin 2\theta}{1+r^2\cos 2\theta}$$

法二：

$$I=\int_0^{\frac{\sqrt{2}}{2}}dy\int_y^{\sqrt{1-y^2}}\frac{xy}{1+x^2-y^2}dx$$

$$=\int_0^{\frac{\sqrt{2}}{2}}\frac{1}{2}y\ln(1+x^2-y^2)\Big|_y^{\sqrt{1-y^2}}dy$$

$$=\frac{1}{2}\int_0^{\frac{\sqrt{2}}{2}}y\ln(2-2y^2)dy=-\frac{1}{8}\int_0^{\frac{\sqrt{2}}{2}}\ln(2-2y^2)d(2-2y^2)$$

$$=-\frac{1}{8}(2-2y^2)\ln(2-2y^2)\Big|_0^{\frac{\sqrt{2}}{2}}+\frac{1}{8}\int_0^{\frac{\sqrt{2}}{2}}(2-2y^2)\cdot\frac{-4y}{2-2y^2}dy$$

$$=\frac{1}{4}\ln 2-\frac{1}{4}y^2\Big|_0^{\frac{\sqrt{2}}{2}}=\frac{1}{4}\ln 2-\frac{1}{8}.$$

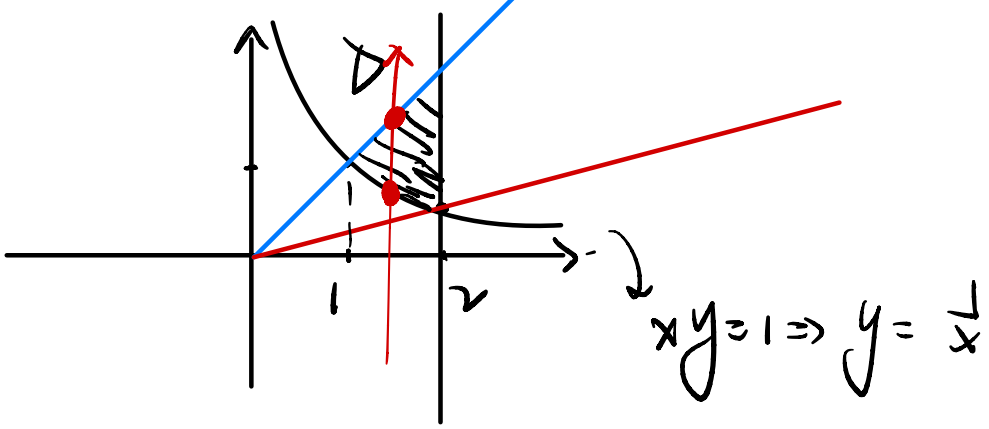
$$\frac{-4y}{2-2y^2}$$

【例6】$I=\int_{\arctan\frac{1}{4}}^{\frac{\pi}{4}}\mathrm{d}\theta\int_{\frac{1}{\sqrt{\sin\theta\cos\theta}}}^{\frac{2}{\cos\theta}}r^2\cos\theta\mathrm{d}r=$_____.

{ 直角坐标
极坐标 顺序

$r=\frac{2}{\cos\theta}\Rightarrow r\cos\theta=2\Rightarrow x=2$

$r=\frac{1}{\sqrt{\sin\theta\cos\theta}}\Rightarrow xy=1$



$$I=\iint_D x\,dx\,dy=\int_1^2 dx\int_{\frac{1}{x}}^{x}x\,dy$$

$$=\int_1^2 x\left(x-\frac{1}{x}\right)dx$$

$$=\left(\frac{1}{3}x^3-x\right)\Big|_1^2=\frac{2}{3}+\frac{2}{3}=\frac{4}{3}$$